

CLOUDNOVA SOLUTIONS

Ransomware Incident Response Playbook

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Threat Actor: DarkCipher | Version 1.0 | April 2026

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INCIDENT SUMMARY	DarkCipher ransomware attack — Monday 7:30 AM 70+ clients affected across 2 data centers
Detection Time	Monday, 7:30 AM — abnormal VM encryption activity observed
Initial Vector	Phishing email to systems engineer containing malicious macro-enabled Excel document
Exploited Vuln	Unpatched third-party monitoring tool on virtual hosts
Threat Actor	DarkCipher — Ransomware-as-a-Service (RaaS) gang
Ransom Demand	15 Bitcoin (BTC)
Clients Affected	70+ SMB clients reporting downtime and inaccessible cloud resources
Scope	Both data centers IaaS VMs Admin dashboards compromised

1. Purpose & Scope

This playbook provides CloudNova Solutions' Incident Response (IR) team with a structured, step-by-step process for detecting, containing, eradicating, and recovering from ransomware attacks. It was developed in direct response to the DarkCipher ransomware attack identified on Monday, April 14, 2026, and is designed to guide all future ransomware incidents.

1.1 Scope

- All CloudNova infrastructure: IaaS virtual machines, storage systems, and network devices across both data centers
- All client-facing services hosted on the CloudNova platform
- All internal systems, endpoints, and administrative tools used by CloudNova staff
- Third-party tools and monitoring agents integrated into the platform

1.2 Framework Alignment

This playbook is aligned with the following industry-recognized frameworks:

Framework	How It Applies
NIST SP 800-61 Rev. 2	Core IR lifecycle: Preparation, Detection, Containment, Eradication, Recovery, Post-Incident
MITRE ATT&CK	Maps DarkCipher TTPs — Initial Access (T1566 Phishing), Execution (T1204), Encryption (T1486)
CIS Controls v8	Controls 4, 6, 7, 8, 12, 17 — patching, logging, malware defenses, IR management
ISO/IEC 27035	Information security incident management principles and guidelines

2. Incident Key Facts — DarkCipher Attack

The following facts were established during initial analysis of the DarkCipher ransomware attack against CloudNova Solutions:

Category	Detail
Company	CloudNova Solutions — IaaS provider for SMBs, 2 data centers, 150+ clients
Attack Date/Time	Monday, April 14, 2026 — detected at 7:30 AM
Initial Access Vector	Phishing email sent to a systems engineer containing a macro-enabled Excel document
Vulnerability Exploited	Unpatched third-party monitoring tool running on virtual hosts
Payload	File-encrypting ransomware deployed across virtual machines in both data centers
Ransom Note	Appeared on administrator dashboards; demand: 15 BTC
Threat Actor	DarkCipher — known Ransomware-as-a-Service group
Clients Impacted	70+ clients reporting downtime and inability to access cloud-hosted resources
First Indicators	Abnormal file encryption on VMs; client downtime reports; ransom note on dashboards
Response Team	CISO, System Administrators, Legal Counsel, Customer Relations

3. Roles & Responsibilities

Each member of the CloudNova IR team carries specific responsibilities before, during, and after a ransomware incident.

Role	Owner	Key Responsibilities
CISO	Chief Information Security Officer	Overall IR coordination and decision authority Authorize containment and recovery actions Liaise with executive leadership and board Decide on ransom payment policy (recommend: do not pay) Approve all public and client communications
System Administrators	Senior Sys Admin Team	Identify and isolate affected VMs and network segments Preserve forensic evidence and system logs Execute system restoration from clean backups Patch and remediate the exploited monitoring tool Monitor for re-infection post-recovery
Legal Counsel	Internal/External Legal Team	Assess regulatory breach notification obligations (GDPR, CCPA, etc.) Advise on ransom payment legality and sanctions risk Manage law enforcement engagement (FBI, CISA) Document all IR actions for legal defensibility Review client SLA breach exposure and liability
Customer Relations	Customer Relations Manager	Notify all 70+ affected clients within required timeframes Provide regular status updates throughout the incident Manage inbound client inquiries and escalations Coordinate post-incident service restoration confirmations Support retention efforts and trust rebuilding

4. Incident Response Playbook — Six Phases

CloudNova's ransomware response follows the NIST SP 800-61 Rev. 2 lifecycle, customized for the IaaS environment and DarkCipher attack profile.

Phase 1: Preparation

Ongoing activities that must be in place before any incident occurs.

4.1.1 Technical Readiness

- Maintain an up-to-date asset inventory of all VMs, containers, endpoints, and third-party tools
- Deploy and tune SIEM (e.g., Splunk, Microsoft Sentinel) with alerts for mass file encryption events
- Enforce multi-factor authentication (MFA) on all admin accounts and remote access
- Implement email filtering with attachment sandboxing — block macro-enabled Office documents from external senders
- Establish and test an immutable, offsite backup strategy (3-2-1 rule: 3 copies, 2 media, 1 offsite)
- Patch and update all third-party tools on a defined schedule; track EOL software
- Segment network — isolate IaaS VMs from admin systems and client traffic where possible

4.1.2 Organizational Readiness

- Maintain a current IR contact list (CISO, Sys Admins, Legal, Customer Relations, external IR firm)
- Conduct phishing awareness training quarterly; simulate phishing campaigns
- Run tabletop exercises simulating ransomware scenarios at least annually
- Establish relationships with FBI Cyber Division and CISA before an incident occurs
- Define and document the ransom payment policy — CloudNova's position: do not pay

Phase 2: Detection & Analysis

Identifying that an incident has occurred and understanding its scope.

4.2.1 Indicators of Compromise (IoCs) — DarkCipher Profile

IoC Type	Observed Indicators
File System	Mass file encryption; files renamed with unknown extensions; ransom note files on filesystem
Network	Unusual outbound traffic to unknown IPs; C2 beaconing; lateral movement between VMs
Process	Suspicious macro execution from Excel; vssadmin.exe deleting shadow copies; PowerShell activity
User Activity	Systems engineer account anomalies; off-hours logins; privilege escalation

Admin Dashboard

Ransom note visible on CloudNova admin panels; client access portals unavailable

4.2.2 Immediate Analysis Steps

1. Verify the alert — confirm encryption activity is malicious, not a legitimate backup or process
2. Activate the IR team — page CISO, Sys Admin lead, Legal, and Customer Relations simultaneously
3. Assign a dedicated Incident Commander (IC) — CISO assumes this role by default
4. Begin evidence collection — snapshot VM states, preserve logs before any changes
5. Determine scope — how many VMs affected? Which clients? Both data centers?
6. Identify Patient Zero — trace back to the phishing email and the systems engineer's endpoint
7. Document timeline — establish chronological order of events from initial access to detection
8. Assess data exfiltration risk — determine whether DarkCipher exfiltrated data before encrypting

Phase 3: Containment

Stopping the spread of ransomware while preserving evidence and minimizing client impact.

4.3.1 Short-Term Containment (Within First Hour)

- Isolate affected VMs immediately — disconnect from network without powering down (to preserve memory)
- Block outbound traffic from affected segments at the firewall level
- Disable the compromised systems engineer account and all associated credentials
- Block DarkCipher-associated IP addresses and domains at perimeter firewall and DNS
- Notify clients — Customer Relations to send initial incident notification within 2 hours

4.3.2 Long-Term Containment

- Rebuild clean replacement VMs in isolated environment for priority clients
- Revoke and rotate all admin credentials, API keys, and service account tokens
- Disable or uninstall the vulnerable third-party monitoring tool across all hosts
- Implement enhanced monitoring on all remaining VMs for signs of lateral spread
- Preserve forensic images of encrypted VMs for legal and investigation purposes

Phase 4: Eradication

Removing all traces of the attacker from the environment.

- Identify and remove all DarkCipher malware artifacts from affected and potentially affected systems
- Scan all hosts with updated anti-malware tools; use known DarkCipher IoC signatures
- Patch the third-party monitoring tool vulnerability or replace the tool entirely
- Audit all VMs not yet confirmed as affected — treat entire environment as potentially compromised
- Verify no persistence mechanisms remain: scheduled tasks, registry keys, backdoors, rogue accounts
- Conduct full log review to ensure no attacker footholds remain in either data center
- Coordinate with external IR firm or MSSP to validate eradication before recovery begins

Phase 5: Recovery

Restoring systems to normal operation safely and systematically.

4.5.1 Recovery Priority Order

Priority	System/Service	Recovery Action
1	Admin Dashboards	Rebuild from clean image; restore monitoring capability for IR

		team
2	High-Impact Clients	Restore top-tier client VMs from last known clean backup
3	Remaining Clients	Systematic restore of all 70+ affected client environments
4	Internal Systems	Restore internal tools after client services are confirmed stable
5	Monitoring Tool	Deploy patched or replacement monitoring solution last

4.5.2 Recovery Checklist

- Restore only from backups created prior to the compromise date — verify backup integrity first
- Test restored systems in isolated environment before returning to production
- Confirm with each affected client that their services are fully operational
- Monitor all restored systems intensively for 72 hours post-recovery
- Do not decrypt using attacker-provided key — rebuilding from backup is always preferred

Phase 6: Post-Incident Activity

Learning from the incident to prevent recurrence and improve defenses.

4.6.1 Lessons Learned Meeting

- Schedule within 2 weeks of full recovery — mandatory attendance from all IR team members
- Review full incident timeline: detection gap, response speed, containment effectiveness
- Identify what controls failed: email filtering, patch management, network segmentation
- Update this playbook based on findings

4.6.2 Root Cause Remediation

- Mandatory phishing simulation training for all staff — focus on macro-enabled documents
- Implement strict patch management SLA: critical vulnerabilities patched within 48 hours
- Review and strengthen third-party vendor security assessment process
- Deploy application allowlisting on admin endpoints
- Enable macro execution blocking via Group Policy for all Office applications

4.6.3 Regulatory & Legal Actions

- File required breach notifications with relevant data protection authorities
- Notify law enforcement: submit report to FBI IC3 and CISA
- Review and update client SLAs and incident notification commitments
- Retain all forensic evidence per legal hold requirements (minimum 3 years)

5. Communication Plan

Audience	Who Communicates	Timing	Key Message
Internal Staff	CISO	Within 1 hour	Incident confirmed; IR activated; do not discuss externally
Affected Clients	Customer Relations	Within 2 hours	Service disruption acknowledged; restoration in progress
All 150+ Clients	Customer Relations	Within 4 hours	Precautionary notice; investigation underway
Executive / Board	CISO	Within 2 hours	Full briefing on scope, impact, response actions
Law Enforcement	Legal Counsel	Within 24 hours	FBI IC3 report; CISA notification
Regulators	Legal Counsel	Per regulation	GDPR: 72 hours; CCPA as applicable
Media (if needed)	CISO + Legal	Only if required	Prepared statement; no operational details

Ransom Payment Policy: CloudNova Solutions does not pay ransoms. Payment does not guarantee decryption, funds criminal enterprises, and may violate OFAC sanctions. All ransom demands are to be reported to Legal Counsel and law enforcement immediately.

6. Recommended Security Improvements

Based on the DarkCipher incident analysis, the following improvements are prioritized:

#	Improvement	Priority	Implementation Action
1	Email Security Enhancement	CRITICAL	Block macro-enabled Office docs from external email; deploy sandbox attachment analysis
2	Patch Management Program	CRITICAL	Critical vulnerabilities patched within 48 hours; track all third-party tool versions
3	MFA Enforcement	HIGH	Mandatory MFA on all admin accounts, VPN access, and cloud management portals
4	Immutable Backup Strategy	HIGH	Implement 3-2-1 backup rule with offsite immutable copies; test restores quarterly
5	Network Segmentation	HIGH	Isolate client VMs from admin systems; implement zero-trust network access
6	Security Awareness Training	MEDIUM	Quarterly phishing simulations; mandatory training upon failure; focus on macro threats
7	EDR Deployment	MEDIUM	Deploy Endpoint Detection & Response on all admin endpoints and engineer workstations
8	Third-Party Risk Program	MEDIUM	Security review before deploying any third-party tool; continuous monitoring of vendor CVEs

7. Appendix — Quick Reference

7.1 Incident Severity Levels

Level	Name	Criteria
P1 — Critical	Active Ransomware	Encryption in progress; multiple systems affected; client data at risk — THIS INCIDENT
P2 — High	Suspected Compromise	Single system affected; threat contained; no confirmed data loss
P3 — Medium	Phishing / Malware	Malicious email opened; no payload executed; endpoint isolated
P4 — Low	Security Alert	Suspicious activity flagged; investigation required; no confirmed threat

7.2 IR Team Emergency Contacts

Role	Name	Phone	Escalation
CISO / Incident Commander	[Insert Name]	[Insert]	Executive Team
Lead System Administrator	[Insert Name]	[Insert]	CISO
Legal Counsel	[Insert Name]	[Insert]	CISO
Customer Relations Manager	[Insert Name]	[Insert]	CISO
External IR Firm	[Insert Vendor]	[Insert]	CISO
FBI Cyber Division	1-855-292-3937	24/7 Hotline	Legal Counsel
CISA	1-888-282-0870	24/7 Hotline	Legal Counsel

7.3 Document Control

Version	Date	Author	Changes
1.0	April 2026	CISO / IR Team	Initial version — developed in response to DarkCipher incident
1.1	[Next Review]	[Name]	Post-incident lessons learned updates

This playbook must be reviewed and updated after every incident and at minimum annually. All team members must acknowledge receipt and understanding of this document.